Digital System Design [CS330]

→ Motivation
→ Applications
→ Augustus, George Boole / Logic & Mathematics

• Digital Logic:

→ Basic Logic gates → OR, AND, NOT
→ 2 state operation → True or False operation.

★ 1 → High o/p (↳ Positive Logic) 0 → High o/p (↳ Negative Logic)

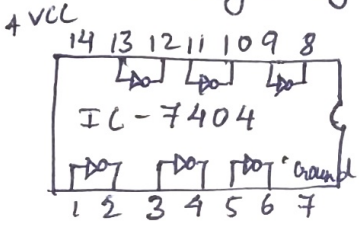
★ NOT gate [Inverter]:



↳ Logic symbol

A	$y=\overline{A}$
0	1
1	0

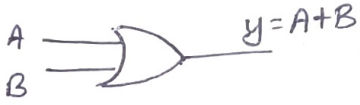
→ Truth table



→ Pinout diagrams.

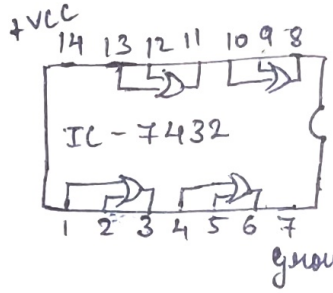
★ OR-gate:

Logic symbol:

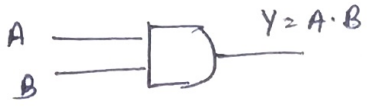


Truth table:

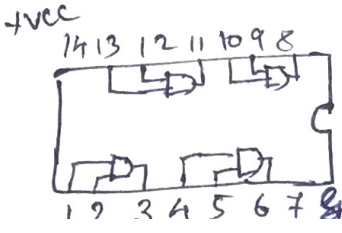
A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1



★ AND gate:



A	B	Y = A·B
0	0	0
0	1	0
1	0	0
1	1	1



②

* Ex-OR gate :

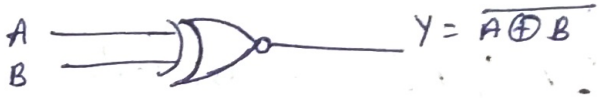


A	B	Y = A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

$Y = \bar{A}B + A\bar{B}$
$= A \oplus B$

IC - 7486

* Ex-NOR gate :



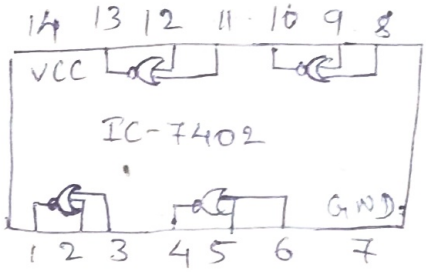
A	B	$Y = \overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

$Y = \bar{A}\bar{B} + AB$
$= \overline{A \oplus B}$

IC - 4077

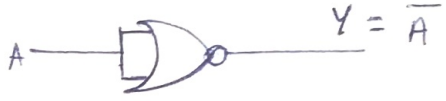
* Universal gates → NAND, NOR :

* NOR gate :

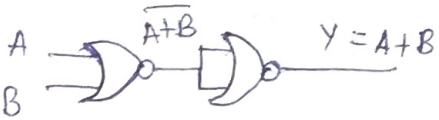


A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

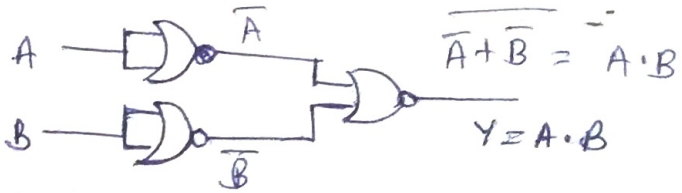
(i) NOT gate :



(ii) OR gate :



(iii) AND gate :

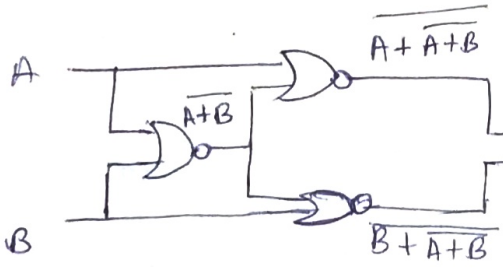


(iv) X-OR gate :



(iv) EX-NOR gate:

$A \oplus B = \bar{A}B + \bar{B}A$



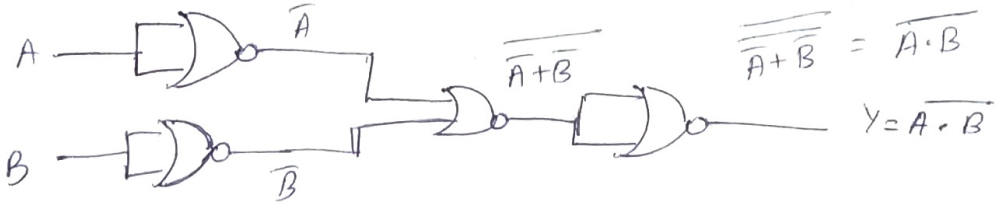
$Y = \overline{A \oplus B}$

$\bar{A} + A + B + B + A + B = A + B + (A \cdot B)$

$= A + \bar{A} \cdot \bar{B} + B + \bar{A} \cdot \bar{B}$

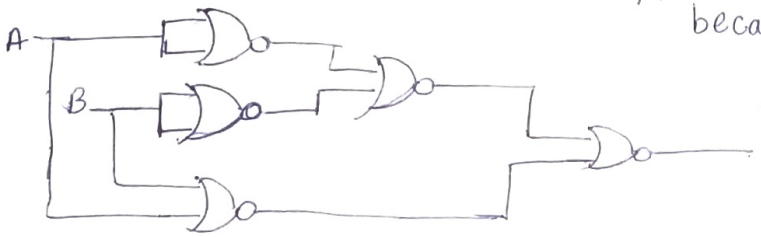
$= \bar{A} \cdot \bar{B} (A + B)$

(v) NAND gate:



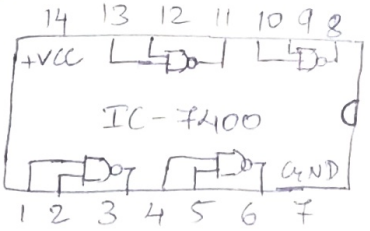
(vi) Ex-OR gate:

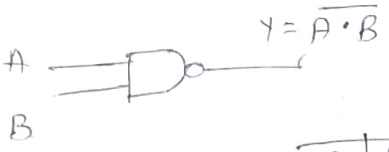
This is the circuit diagram because of propogation delay.



$Y = A \oplus B$

★ NAND gate:



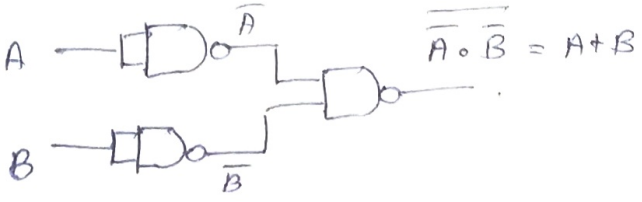


A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

(i) NOT gate:



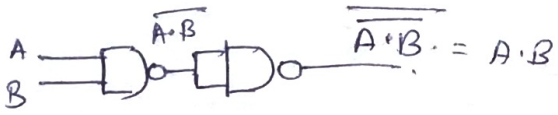
(ii) OR gate:



④

(iii) AND gate:

A	$\bar{A}$	$A \cdot \bar{A}$
0	1	0
1	0	0

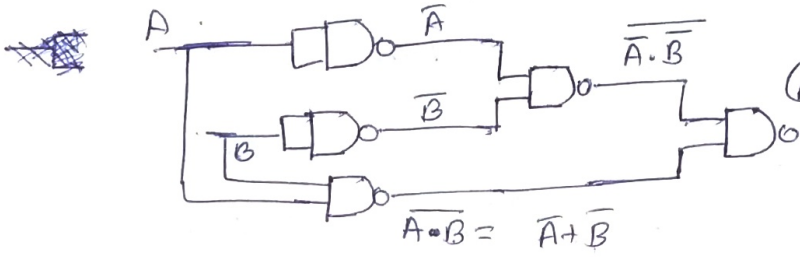


(iv) ExNOR gate: $\bar{A}B + A\bar{B} = \overline{\overline{AB} \cdot \overline{A\bar{B}}}$ $\bar{A} \cdot \bar{B} = \overline{(A+B)}$

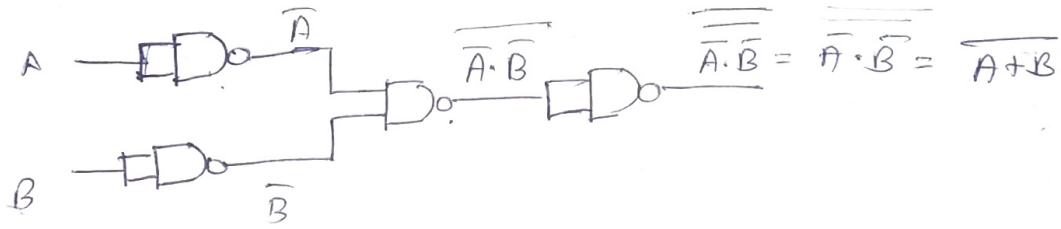
$A\bar{A} + A\bar{B} + \bar{A}B + B\bar{B}$

$(A+B)(\bar{A}+\bar{B}) = A\bar{B} + \bar{A}B$

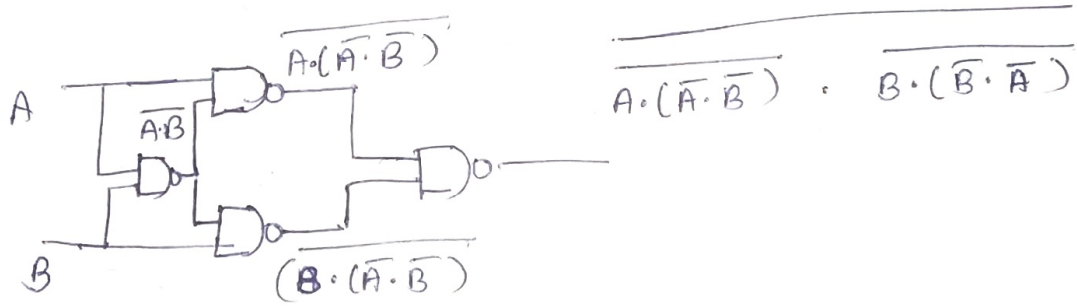
$= A \oplus B$



(v) NOR gate:



(vi) Ex-OR gate:



$= \overline{\overline{A \cdot (\bar{A} \cdot \bar{B})}} + \overline{\overline{B \cdot (\bar{A} \cdot \bar{B})}}$

$= A \cdot (\bar{A} \cdot \bar{B}) + B \cdot (\bar{A} \cdot \bar{B})$

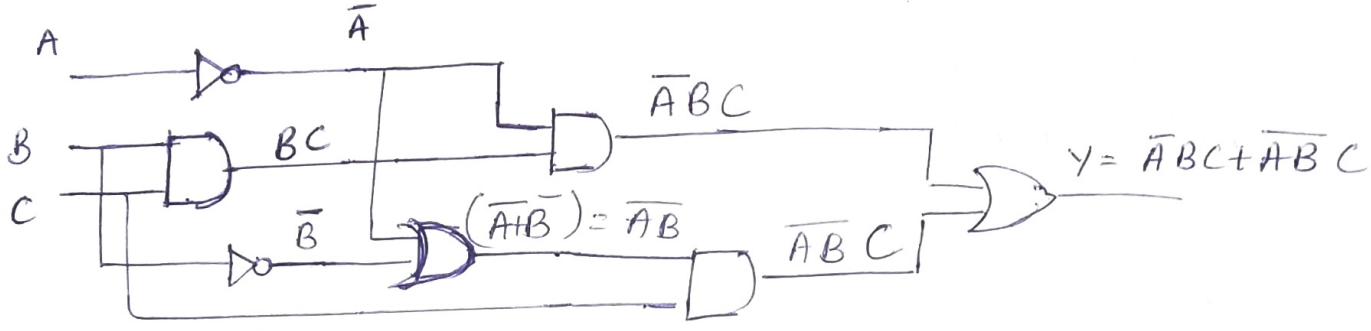
$= (\bar{A} \cdot \bar{B})($

1.) Write the circuit diagram for

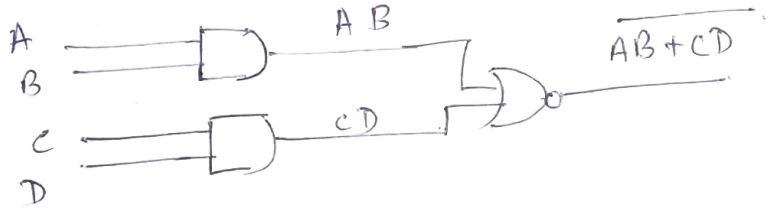
(i) $Y = \bar{A}BC + \bar{A}\bar{B}C$.

(ii) $Y = \overline{AB + CD}$

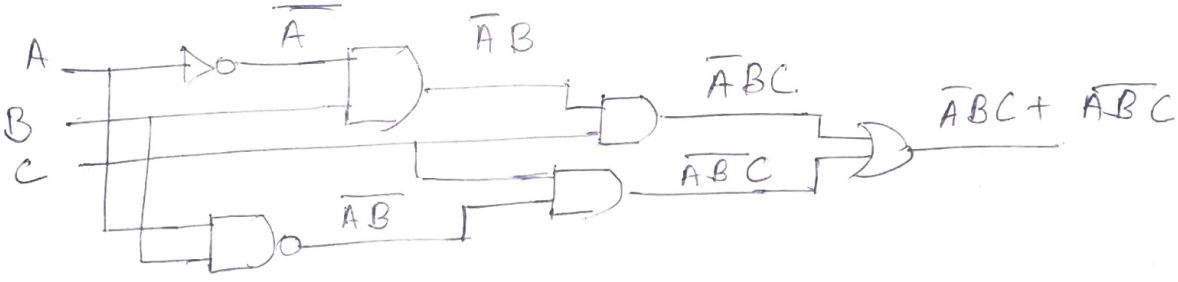
Ans: (i) $Y = \bar{A}BC + \bar{A}\bar{B}C$



(ii) $Y = \overline{AB + CD}$

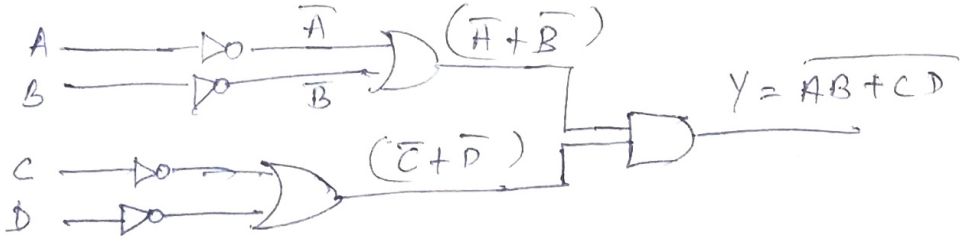


(i)



(ii) $Y = \overline{AB+CD} = (\overline{AB}).(\overline{CD}) = (\bar{A}+\bar{B})(\bar{C}+\bar{D})$

$\neq \overline{A}+\overline{B}$



• Boolean Laws and Theorems:

* Commutative Law:

$$A+B = B+A \quad , \quad AB = BA$$

* Associative law:

$$A+(B+C) = (A+B)+C$$

$$(AB)C = A(BC)$$

* Distributive law:

$$A(B+C) = AB+AC$$

* OR operations:

$$A+0 = A$$
$$A+A = A$$
$$A+1 = 1$$
$$A+\bar{A} = 1$$

* AND operations:

$$A \cdot 0 = 0$$
$$A \cdot A = A$$
$$A \cdot 1 = A$$
$$A \cdot \bar{A} = 0$$

* Double inversion:

$$\bar{\bar{A}} = A$$

* De-Morgan's theorem:

$$\overline{A+B} = \bar{A} \cdot \bar{B}$$
$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

* Duality theorem:

AND - OR

OR - AND

Complementing only 0 or 1

* Covering rule (Absorption rule).

$$AB + A\bar{B} = A$$

$$(A+B)(A+\bar{B}) = A$$

★ Consensus rule (theorem):

$AB + \bar{A}C + BC = AB + \bar{A}C$

$(A+B)(\bar{A}+C)(B+C) = (A+B)(\bar{A}+C)$

If any unique boolean term is not present, then it is redundant

2.) $\overline{AB + A\bar{B}}$

$= (\bar{A}+\bar{B})(\bar{A}+B)$ [Covering rule]

$= \bar{A}$

★ $A + \bar{A}B = A + B$

$AB + A\bar{B} = A$

$(A+B)(A+\bar{B})$

$= A \cdot A + B\bar{B} + AB + A\bar{B}$

$= A + (AB + A\bar{B})$

$= A + A = A$.

$\overline{(A+B)(A+\bar{B})} = \overline{(A+B)} + \overline{(A+\bar{B})}$

$= \bar{A}\bar{B} + \bar{A}B = \bar{A} = A$

Problems:

1.) $F = BC + B\bar{C} + BA$

$= BC + B\bar{C} + BA$

$= B + BA$

$= B(1+A)$

$F = B$

2.) $F = AB\bar{C}\bar{D} + AB\bar{C}D + A\bar{B}\bar{C}D + ABCD + A\bar{B}CD + ABC\bar{D} + A\bar{B}C\bar{D}$

$= AB\bar{C}[D+\bar{D}] + A\bar{B}D[C+\bar{C}] + AC\bar{D}[B+\bar{B}] + ABC[D+\bar{D}]$

$= AB\bar{C} + A\bar{B}D + AC\bar{D} + ABC$

$= AB[\bar{C}+C] + A\bar{B}D + AC\bar{D}$

$= AB + A\bar{B}D + AC\bar{D}$ $= A(B + \bar{B}D) + AC\bar{D}$

$= A(B+D) + AC\bar{D}$

(8)

3.) $\overline{A(\bar{B}\bar{C}+BC)} = \bar{A}+(B+C)(\bar{B}+\bar{C})$

$LHS = \overline{A(\bar{B}\bar{C}+BC)}$

$= \bar{A} + \overline{(\bar{B}\bar{C}+BC)}$ [De morgan's theorem]

$= \bar{A} + (\overline{(\bar{B}\bar{C})} \cdot \overline{(BC)})$

$= \bar{A} + ((B+C)(\bar{B}+\bar{C}))$ = RHS

2.) $F = A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}CD + ABCD + A\bar{B}CD + AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}C\bar{D}$

$= A\bar{B}\bar{C}[D+\bar{D}] + A\bar{B}D[C+\bar{C}] + ABC[D+\bar{D}] + A\bar{B}C\bar{D}$ (1, 1, 1)

$= A\bar{B}\bar{C} + A\bar{B}D + ABC + A\bar{B}C\bar{D}$

$= AB(C+\bar{C}) + A\bar{B}(D+C\bar{D})$ (1)

$= AB + A\bar{B}(C+D)$

$= AB + A\bar{B}C + A\bar{B}D$

$= A(B+\bar{B}C) + A\bar{B}D$

$= A(B+C) + A\bar{B}D$

$= AB + AC + A\bar{B}D$

$= A(B+\bar{B}D) + AC$

$= A(B+D) + AC$

$= AB + AC + AD$

4.) $A(\bar{A}+C)(\bar{A}B+C)(\bar{A}BC+\bar{C})=0$.

Proof:

$$LHS = A(\bar{A}+C)(\bar{A}B+C)(\bar{A}BC+\bar{C})$$
$$= A\bar{A}+AC(\bar{A}B+C)(\bar{A}BC+\bar{C})$$
$$= (A\bar{A}BC+AC\cdot C)(\bar{A}BC+\bar{C}).$$
$$= AC(\bar{A}BC+\bar{C})$$
$$= (A\bar{A}BC\cdot C)+(AC\cdot\bar{C})$$
$$= 0 = RHS$$

5.) $1 = (A+B)\left(\overline{(\bar{A}+(B+C))}+\bar{A}(B+C)\right)$

$$RHS = (A+B)\left(\left(\bar{A}\cdot(\overline{\bar{B}+\bar{C}})\right)+\bar{A}(B+C)\right)$$
$$= (A+B)\left(\left(A\cdot(\bar{B}\cdot\bar{\bar{C}})\right)+\bar{A}(B+C)\right)$$
$$= (A+B)(ABC+\bar{A}B+\bar{A}C)$$
$$= (A+B)(B(AC+\bar{A})+\bar{A}C)$$
$$= (A+B)(B(\bar{A}+C)+\bar{A}C)$$
$$= (A+B)(\bar{A}B+BC+\bar{A}C)$$
$$= A\cdot\bar{A}B+ABC+A\bar{A}C+\bar{A}B\cdot B+B\cdot BC+B\cdot\bar{A}C.$$
$$= ABC+\bar{A}B+BC+\bar{A}BC$$
$$= BC(A+\bar{A})+\bar{A}B+BC$$
$$= BC+\bar{A}B$$
$$=$$

(10)

5) $(A+B)\overline{(\bar{A}+\bar{B}+\bar{C})} + \bar{A}(B+C)$

$= (A+B)(\bar{\bar{A}} \cdot \overline{(\bar{B}+\bar{C})}) + \bar{A}(B+C)$

$= (A+B)(A \cdot (\bar{\bar{B}} \cdot \bar{\bar{C}})) + \bar{A}(B+C)$

$= (A+B)(ABC) + \bar{A}(B+C)$

$= ABC + ABC + \bar{A}(B+C)$

$= ABC + \bar{A}B + \bar{A}C$

$= B(C\bar{A}+AC) + \bar{A}C$

$= (\bar{A}+C)B + \bar{A}C$

$= \bar{A}B + BC + \bar{A}C \quad = \bar{A}(B+C)$

$= \bar{A}(B+C) + BC$

5) $Y = (A+B)\overline{((A+(\bar{B}+\bar{C}))} + \bar{A}(B+C)$

$= (A+B)(\bar{A} \cdot \overline{(\bar{B}+\bar{C})}) + \bar{A}(B+C)$

$= (A+B)(\bar{A} \cdot (\bar{\bar{B}} \cdot \bar{\bar{C}})) + \bar{A}(B+C)$

$= (A+B)(\bar{A} \cdot BC) + \bar{A}B + \bar{A}C$

$= (A+B)Y$

$= A\bar{A}BC + B\bar{A}BC + \bar{A}B + \bar{A}C$ (0)

$= \bar{A}BC + \bar{A}B + \bar{A}C$

$= \bar{A}B(C+1) + \bar{A}C$ (1)

$= \bar{A}B + \bar{A}C$

5.) $Y = (A+B)\left(\overline{(A+(\bar{B}+\bar{C}))} + \bar{A}(B+C)\right)$

$= (A+B)\left(\bar{A}\cdot\overline{(\bar{B}+\bar{C})} + \bar{A}(B+C)\right)$

$= (A+B)(\bar{A}BC + \bar{A}B + \bar{A}C)$

$= (A+B)(\bar{A}B(C+1)' + \bar{A}C)$

$= (A+B)(\bar{A}B + \bar{A}C)$

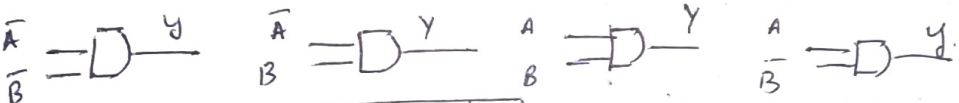
$= A\cdot\bar{A}B + A\cdot\bar{A}C + \bar{A}B\cdot B + \bar{A}BC$ ($A\cdot\bar{A}=0$)

$= \bar{A}B + \bar{A}BC$

$= \bar{A}B(C+1)' = \bar{A}B$

- **Boolean Logic:**

→ **Sum of Products:**

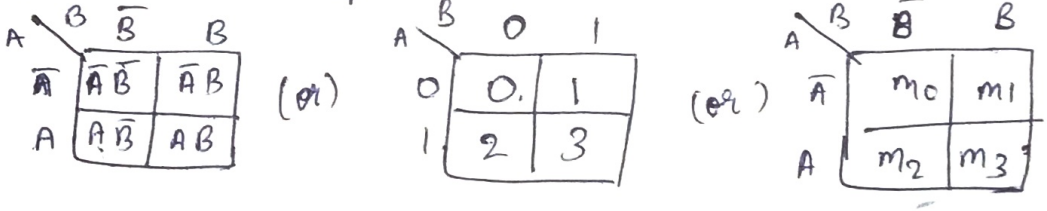


A	B	Y	
0	0	$\bar{A}\bar{B}$	m_0
0	1	$\bar{A}B$	m_1
1	0	$A\bar{B}$	m_2
1	1	AB	m_3

★ Fundamental product
★ Min terms
★ Cannonical sum form – take the output where it is high.

K-map

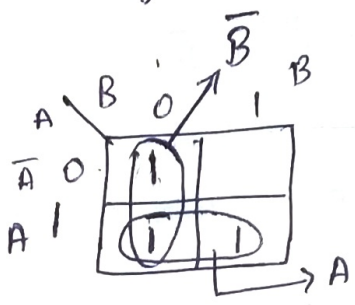
① 2 variable K map



(2) (iii) Three variable K-map

AB \ C	$\bar{C}$	C
$\bar{A}\bar{B}$	0	1
$\bar{A}B$	2	3
$A\bar{B}$	6	7
AB	4	5

Two consecutive 1's – Pair
Four – Quad (Looks like square or end to end rep)
Eight – Octet



$\therefore A+\bar{B}$

$Y = A\bar{B} + AB + A\bar{B}$

$$\overline{(A+B)(\bar{A}(\bar{B}+\bar{C}))} + \bar{A}(B+C)$$

$$= (A+B)(A+\overline{(\bar{B}+\bar{C})}) + \bar{A}(B+C)$$

$$= (A+B)(A+(B\cdot C)) + \bar{A}(B+C)$$

$$= (A\cdot A + ABC + AB + BC) + \bar{A}(B+C)$$

$$= A(1+BC+B) + BC + \bar{A}B + \bar{A}C$$

$$= A + BC + \bar{A}B + \bar{A}C$$

$$= A + \bar{A}B + BC + \bar{A}C$$

$$= A + B + BC + \bar{A}C$$

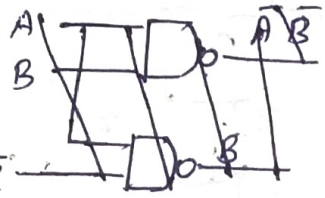
$$= A + C + B + BC$$

$$= A + B + C$$

1.) Simplify $Y = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C$.

AB \ C	$\bar{C}$	C
$\bar{A}\bar{B}$	1	1
$\bar{A}B$	0	1
AB	0	0
$A\bar{B}$	0	1

$\bar{A}\bar{B} + \bar{A}C + \bar{B}C = Y$.



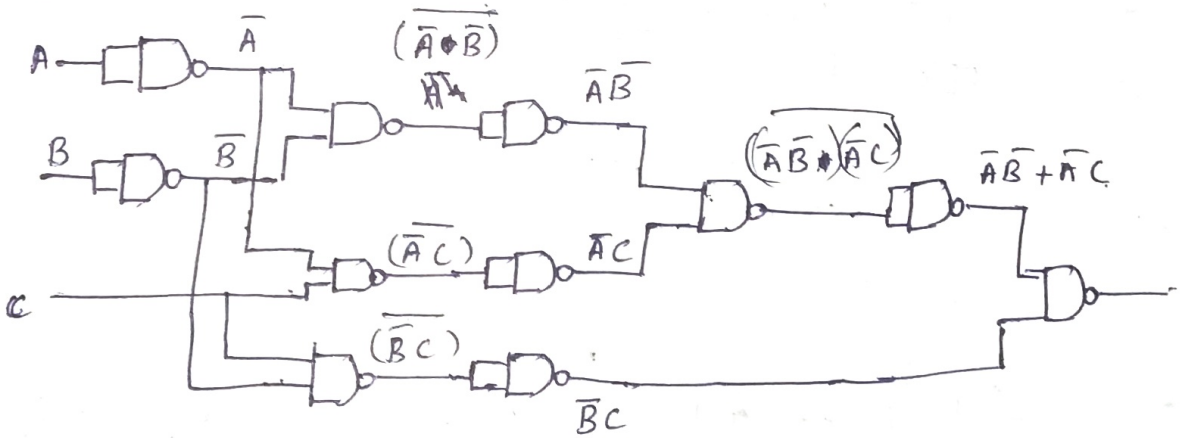
2.) Simplify $F(A, B, C, D) = \bar{A}\bar{B}\bar{C} + \bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}\bar{C}$

Consider both $\bar{A}\bar{B}\bar{C}\bar{D}$ and $\bar{A}\bar{B}\bar{C}D$.

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	1	1	0	1
$\bar{A}B$	0	0	0	1
AB	0	0	0	0
$A\bar{B}$	1	1	0	1

③ $Y = \bar{B}\bar{C} + \bar{B}\bar{D} + \bar{A}C\bar{D}$

b) Logic diagram:

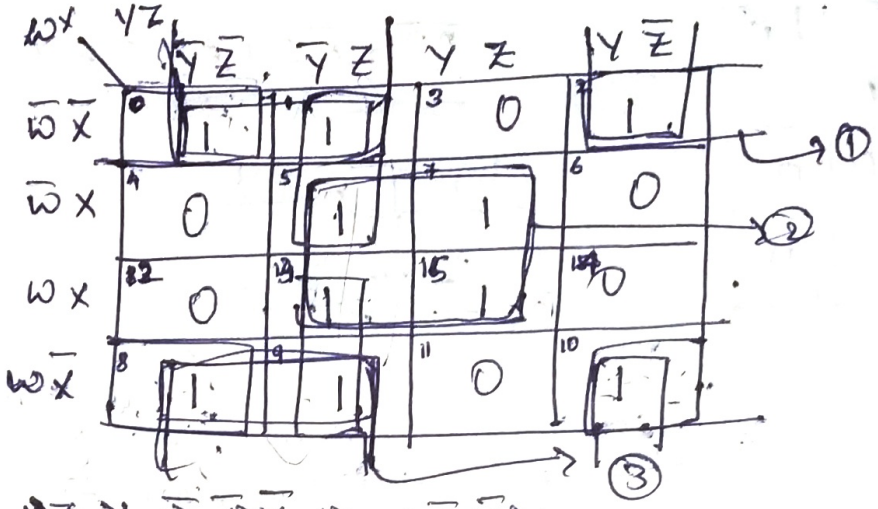


If different answers are gotten, choose the one with lower gates.

Redundant groups are not to be included.

(14)

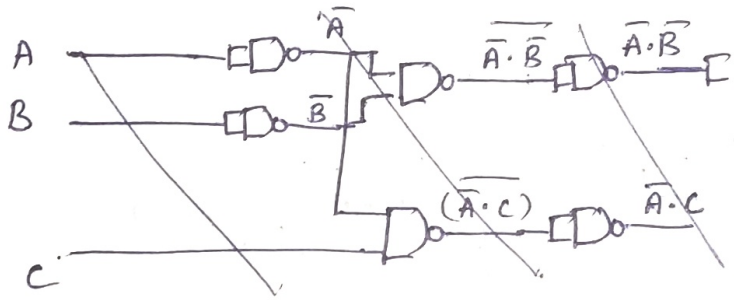
Simplify $F(w, x, y, z) = \Sigma m(0, 1, 2, 5, 7, 8, 9, 10, 13, 15)$

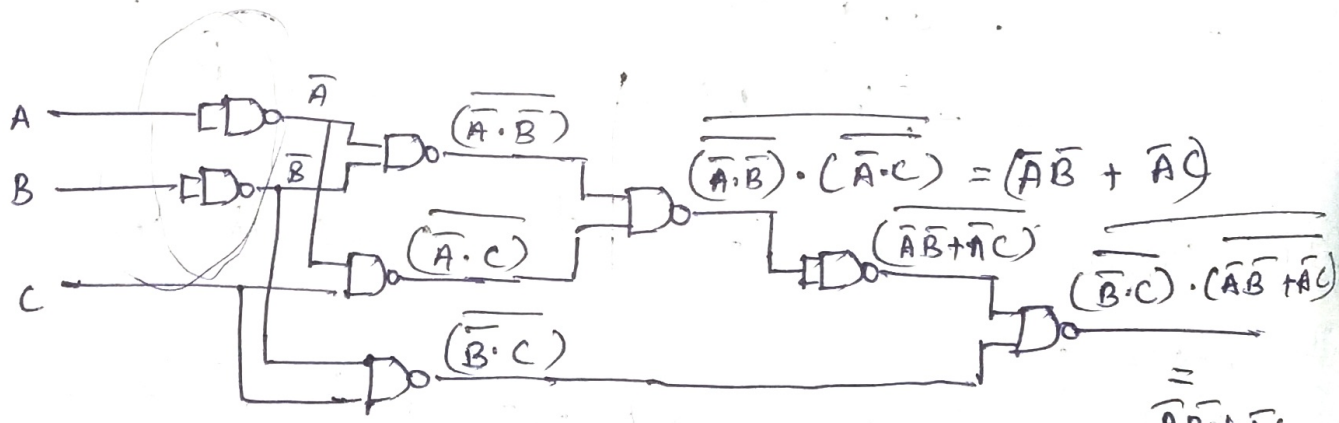


~~$xz + \bar{w}\bar{x}\bar{y} + w\bar{x}\bar{y}$~~

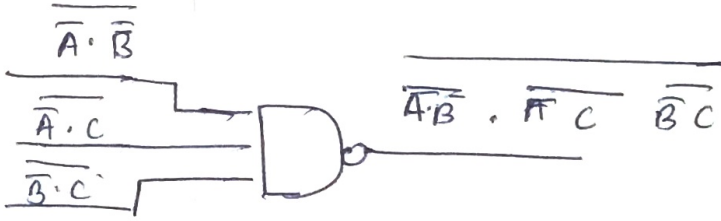
$xz + \bar{x}\bar{y} + \bar{z}\bar{x}$

$Y = \bar{A}\bar{B} + \bar{A}C + \bar{B}C$

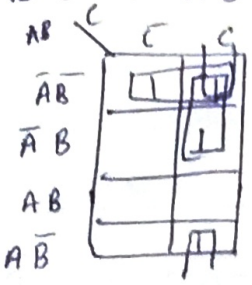




$= \bar{A}\bar{B} + \bar{A}C + \bar{B}C$



$y = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C$



$Y = \bar{A}\bar{B} + \bar{A}C + \bar{B}C$

1) Convert the OR to AND form

Apply double complent & de Morgan's theorem for solving.

Design NAND logic for the equation.

$$y = (\bar{A}B + C)\bar{D}$$

$$\bar{y} = \overline{(\bar{A}B + C)\cdot\bar{D}}$$

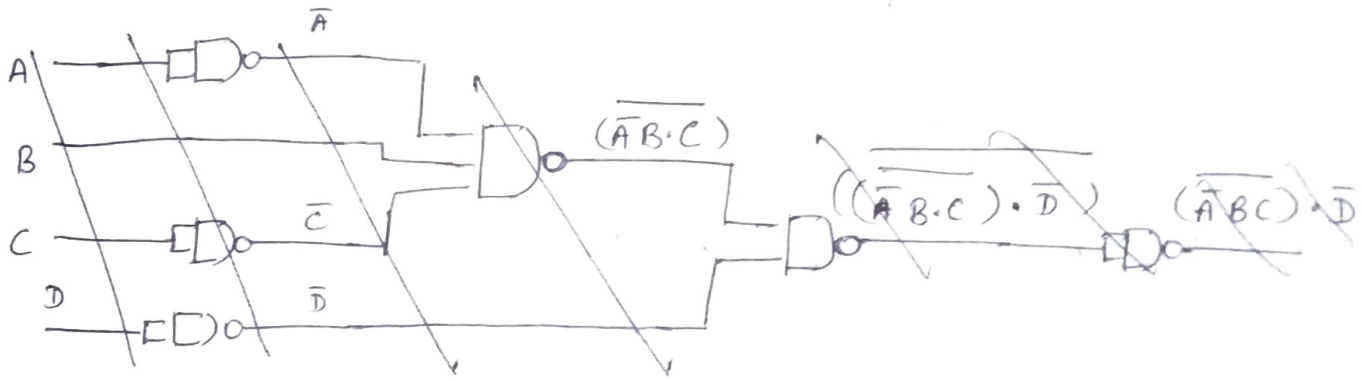
$$= \overline{\overline{(\bar{A}B + C)} + \bar{\bar{D}}}$$

$$= \overline{(\overline{\bar{A}B}\cdot\bar{C}) + D}$$

$$= \overline{(A + \bar{B})\cdot\bar{C} + D}$$

$$= \overline{(\bar{\bar{A}}B\cdot\bar{C})\cdot\bar{D}}$$

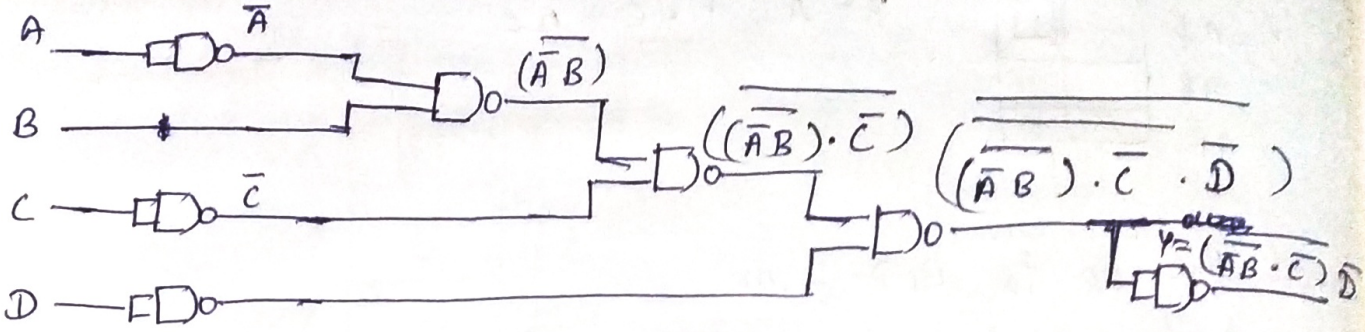
$\overline{A\,B\,C}$





(16)

$$Y = (\overline{\overline{A \cdot B} \cdot \overline{C}}) \cdot \overline{D}$$

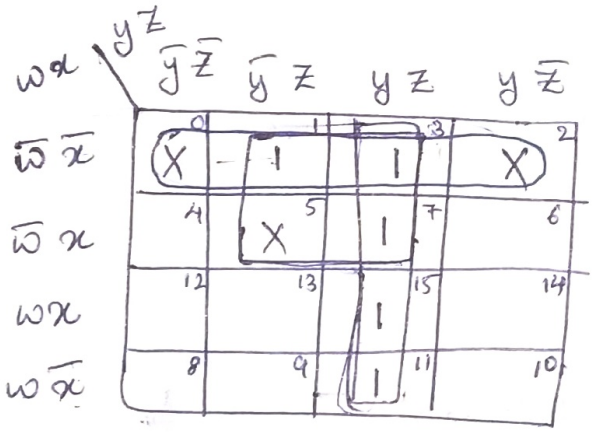


- Don't care condition :

SOP – X – 1
POS – X – 0



$F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ & $d(w, x, y, z) = \Sigma(0, 2, 5)$



$Y = z\bar{w} + yz + \cancel{w\bar{x}}$

~~(or)~~ ~~$Y = \bar{y}\bar{z} + \bar{w}\bar{x}$~~

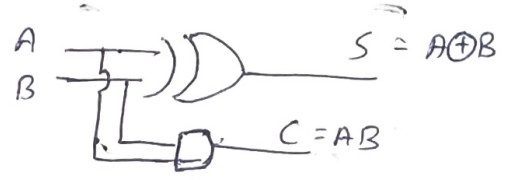
- **Arithmetic building Blocks**
  - → Adder
  - → Subtractor
  - → controller Inverter

① **Half adder :**

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$S = \bar{A}B + A\bar{B} = A \oplus B$

$C = AB$

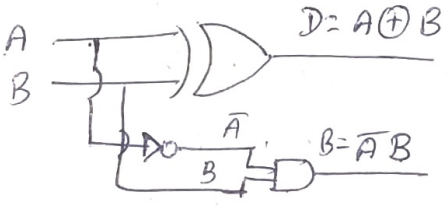


② **Half Subtractor :**

A	B	D	B_{out}
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$D = \bar{A}B + A\bar{B} = A \oplus B$

$B_{out} = \bar{A}B$



- **Product of Sum Method:**
  - ★ In SOP : 1 – A, 0 – $\bar{A}$
  - ★ In POS : 1 – $\bar{A}$, 0 – A

A	B	Y
0	0	M_0
0	1	M_1
1	0	M_2
1	1	M_3

$(A+B)(A+\bar{B})(\bar{A}+B)(\bar{A}+\bar{B})$

★ Fundamental Sum

★ Cannonical product form

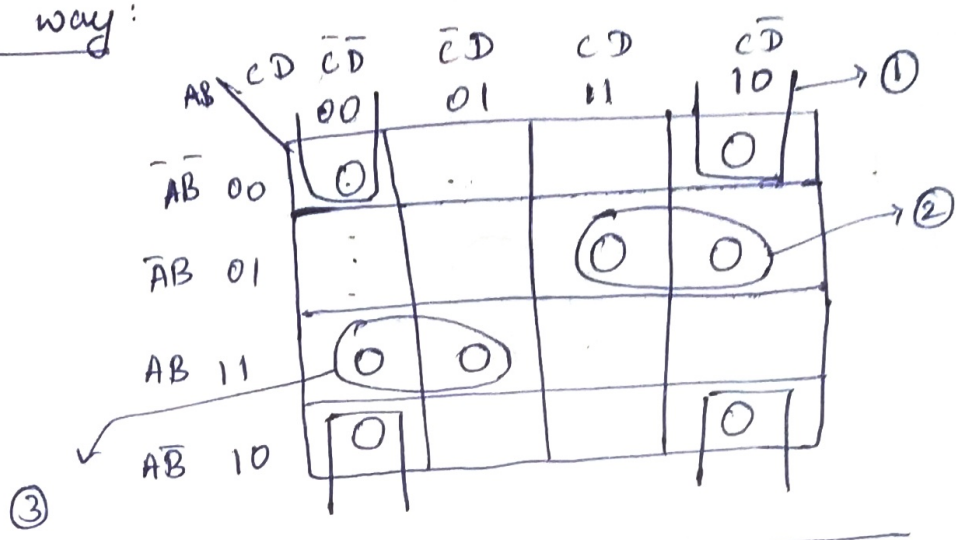
To get POS:

1 - SOP
0 - POS

Ex: $F(A,B,C,D) = \Sigma m(1,3,4,5,9,11,14,15)$

$\overline{F}(A,B,C,D) = \Sigma\Pi(2,6,7,8,10,12,13)$

1st way:



$$\overline{F}(A,B,C,D) = (\bar{B}\bar{D} + \bar{A}BC + AB\bar{C})$$

$$= \overline{(\bar{B}\bar{D})} \cdot \overline{(\bar{A}BC + AB\bar{C})}$$

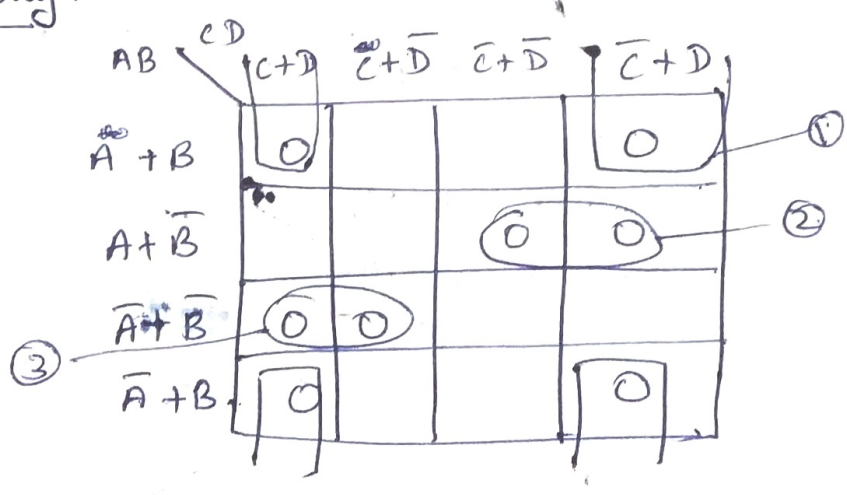
$$= \overline{(\bar{B}\bar{D})} \cdot \overline{(\bar{A}BC)} \cdot \overline{(AB\bar{C})}$$

$$= (\bar{\bar{B}} + \bar{\bar{D}}) \cdot (\bar{\bar{A}}B + \bar{C}) \cdot (\bar{A}\bar{B} + \bar{\bar{C}})$$

$$= (B+D)(\bar{\bar{A}} + \bar{B} + \bar{C}) \cdot (\bar{A} + \bar{B} + C)$$

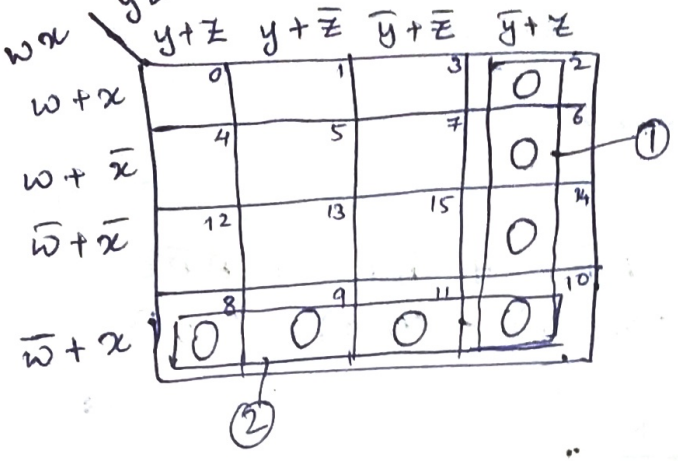
$$= (B+D)(A + \bar{B} + \bar{C})(\bar{A} + \bar{B} + C)$$

2nd way:



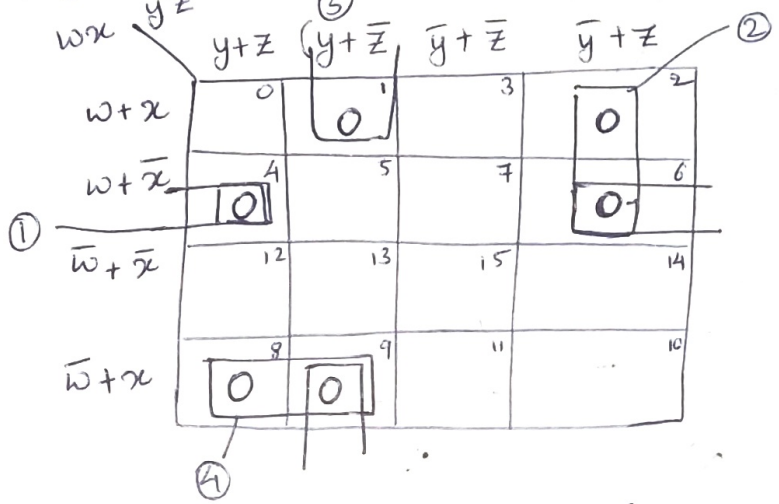
$$\therefore \overline{F}(A,B,C,D) = (B+D)(\bar{A}+\bar{B}+C)(A+\bar{B}+\bar{C})$$

3.) $F(w, x, y, z) = \Pi_M (2, 6, 8, 9, 10, 11, 14)$



$F(w, x, y, z) = (\bar{w} + x)(\bar{y} + z)$

④ $F(w, x, y, z) = \Pi_M (1, 2, 4, 6, 8, 9)$



$(\bar{w} + x + y)(w + \bar{y} + z)(x + y + \bar{z})(w + \bar{x} + z)$

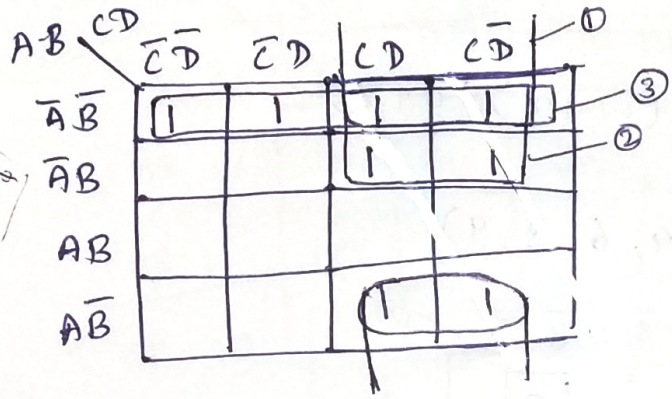
• Quine - McCluskey Method:

Step I: Prime Impliments

Step II: P.I.

Ex: $F(A\,B\,C\,D) = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C$

(0 0 0) (0 0 1) (0 1 1) (1 0 1)



$\bar{B}C + \bar{A}C + \bar{A}\bar{B}$

2 | 14
2 | 7 - 0
2 | 3 - 1
1 - 1

Step I:

No. of 1	A	B	C		AB	C
0	0	0	0	-	0	0 -
1	0	0	1	-	0	- 1
				-	-	0 1
2	0	1	1			
	1	0	1			

Step II:

		0	1	3	5
$\bar{A}\bar{B}$	0, 1	①	1		
$\bar{A}C$	1, 3		1	①	
$\bar{B}C$	1, 5		1		①

$\therefore F(A\,B\,C\,D) = \bar{A}\bar{B} + \bar{A}C + \bar{B}C$

$Y = F(A, B, C, D) = \Sigma m\ (0, 1, 2, 3, 10, 11, 12, 13, 14, 15)$

(0000) (0001) (0010) (0011) (1010) (1011) (1100) (1101) (1110) (1111)

	A B C D
0	0 0 0 0 (0)
1	0 0 0 1 (1)
	0 0 1 0 (2)
2	0 0 1 1 (3)
	1 0 1 0 (10)
	1 1 0 0 (12)
3	1 0 1 1 (11)
	1 1 0 1 (13)
	1 1 1 0 (14)
4	1 1 1 1 (15)

	A B C D
0	0 0 0 – (0, 1)
	0 0 – 0 (0, 2)
1	0 0 – 1 (1, 3)
	~~– 0 – 0 (0, 2, 8, 10)~~
	0 0 1 – (2, 3)
	– 0 1 0 (2, 10)
2	– 0 1 1 (3, 11)
	1 0 1 – (10, 11)
	1 – 1 0 (10, 14)
	1 1 0 – (12, 13)
	1 1 – 0 (12, 14)
3	1 – 1 1 (11, 15)
	1 1 – 1 (13, 15)
	1 1 1 – (14, 15)

	A B C D
0	– 0 0 – (0, 1, 2, 3)
	– 0 0 – (0, 2, 1, 3)
1	– 0 1 – (2, 3, 10, 11)
	– 0 1 – (2, 10, 3, 11)
2	1 – 1 – (10, 11, 14, 15)
	1 – 1 – (10, 14, 11, 15)
	1 1 – – (12, 13, 14, 15)
	1 1 – – (12, 13, 14, 15)

PI :

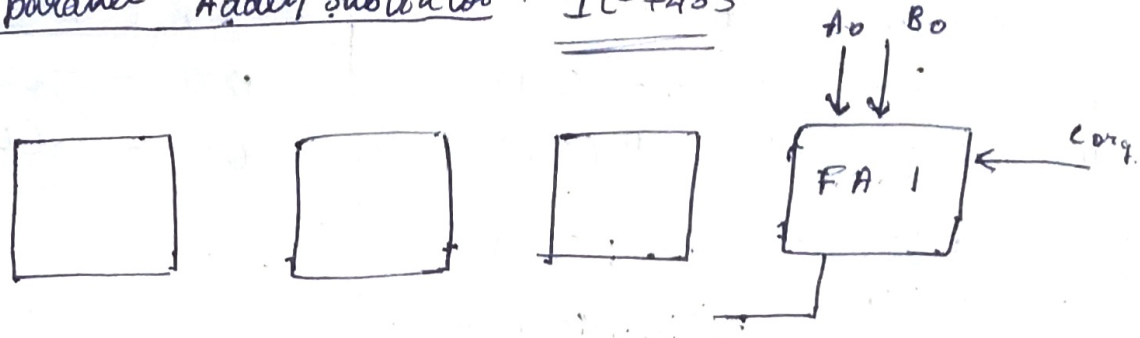
		0	1	2	3	10	11	12	13	14	15
$\bar{A}\bar{B}$	0, 1, 2, 3	⊗	⊗	×	×						
$\bar{B}C$	2, 3, 10, 11			×	×	×	×				
AC	10, 11, 14, 15					⊗	⊗			×	×
AB	12, 13, 14, 15							⊗	⊗	×	×

$\bar{A}\bar{B} + AC + AB$

(or) $\bar{A}\bar{B} + \bar{B}C + AB$

1 (2) (22)

*4 bit parallel Adder/Subtractor : IC-7483



$F(A\,B\,C\,D) = \Sigma_m(0, 1, 3, 7, 8, 9, 11, 15)$

0 - 0000
1 - 0001
3 - 0011
7 - 0111
8 - 1000
9 - 1001
11 - 1011
15 - 1111

Step I

Group	Minterms	A	B	C	D
0	0	0	0	0	0
1	1	0	0	0	1
	8	1	0	0	0
2	3	0	0	1	1
	9	1	0	0	1
3	7	0	1	1	1
	11	1	0	1	1
4	15	1	1	1	1

Step II

	A B C D
0	0 0 0 — (0,1)
	— 0 0 0 (0,8)
1	0 0 — 1 (1,3)
	— 0 0 1 (1,9)
2	1 0 0 — (8,9)
	0 — 1 1 (3,7)
	— 0 1 1 (3,11)
	1 0 — 1 (9,11)
3	— 1 1 1 (7,15)
	1 — 1 1 (9,15)

Step III

Group	m.m	A B C D	
0		— 0 0 — [(0,1),(8,9)]	$\bar{B}\,\bar{C}$
		— 0 0 — [(0,8),(1,9)]	
1		— 0 — 1 [(1,3),(9,11)]	$\bar{B}\,D$
		— 0 — 1 [(1,9),(3,11)]	
2		— — 1 1 [(3,11),(7,15)]	$C\,D$
		— — 1 1 [(3,7),(9,15)]	

	minterms	0	1	3	7	8	9	11	15
$\bar{B}\,\bar{C}$	0, 1, 8, 9	(X)	X			(X)	X		
$\bar{B}\,D$	1, 3, 9, 11		X	X			X	X	
$C\,D$	3, 11, 7, 15			X	(X)			X	(X)

$F = \bar{B}\,\bar{C} + C\,D$

(24)

Design and realize the following using Basic gates by IC 7483

(i) BCD to excess-3 code

(ii) excess-3 code to BCD

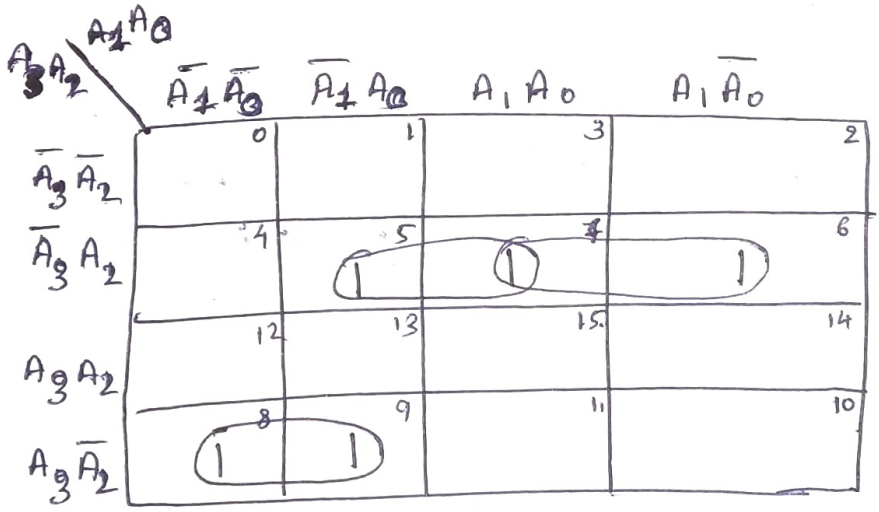
Decimal	BCD A_3 A_2 A_1 A_0	Excess-3 code B_3 B_2 B_1 B_0
0	0 0 0 0	0 0 1 1
1	0 0 0 1	0 1 0 0
2	0 0 1 0	0 1 0 1
3	0 0 1 1	0 1 1 0
4	0 1 0 0	0 1 1 1
5	0 1 0 1	1 0 0 0
6	0 1 1 0	1 0 0 1
7	0 1 1 1	1 0 1 0
8	1 0 0 0	1 0 1 1
9	1 0 0 1	1 1 0 0

1^s complement = 1100

0 1 1 1
− 0 1 0 1
1 0 1 0

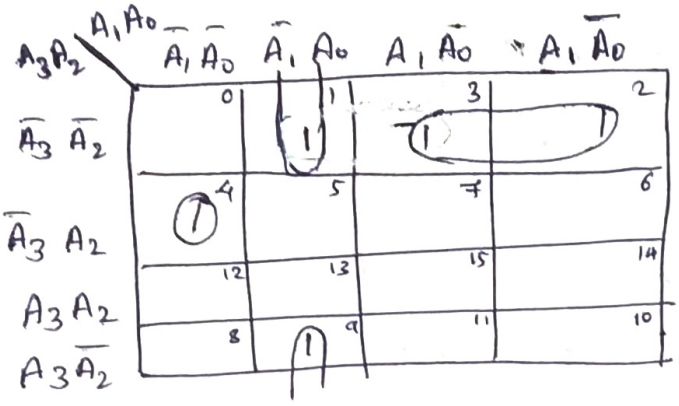
1 1 0 0 1
0 0 1 1
1 1 0 0

self complementary



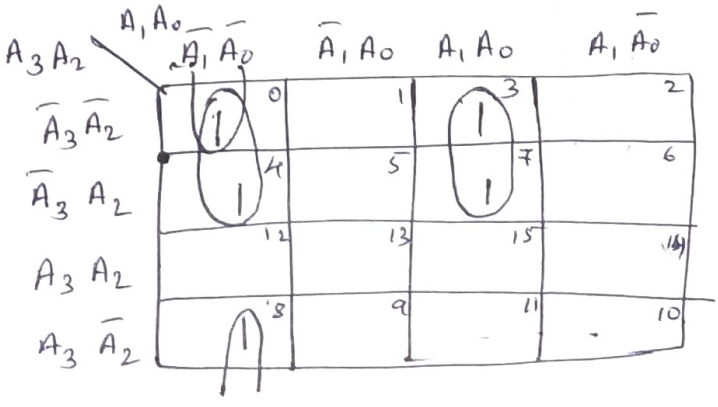
$$B_3 = A_3\bar{A}_2\bar{A}_1 + \bar{A}_3A_2A_0 + \bar{A}_3A_2A_1$$

For B_2 :



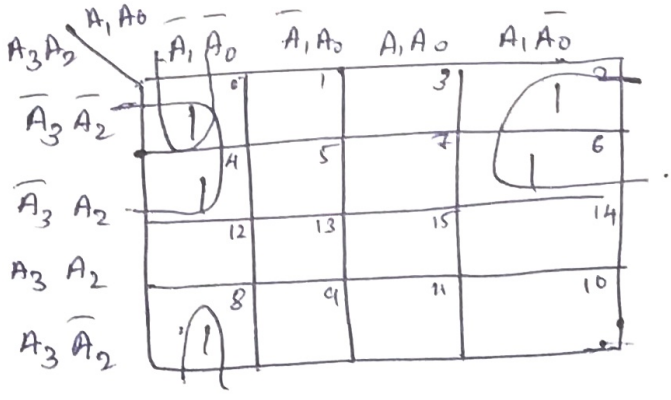
$$B_2 = \bar{A}_3 A_2 \bar{A}_1 \bar{A}_0 + \bar{A}_2 \bar{A}_1 A_0 + \bar{A}_3 \bar{A}_2 A_1$$

For B_1 :



$$B_1 = \bar{A}_3 \bar{A}_1 \bar{A}_0 + \bar{A}_2 \bar{A}_1 \bar{A}_0 + \bar{A}_3 A_1 A_0$$

For B_0 :



$$B_0 = \bar{A}_3 \bar{A}_0 + \bar{A}_2 \bar{A}_1 \bar{A}_0$$

(vi) Excess 3 to BCD

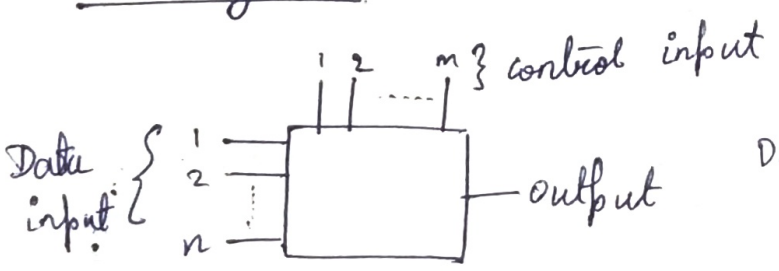
For IC 7483
C_{in} = 0 - Add 3
= 1 - Subtract

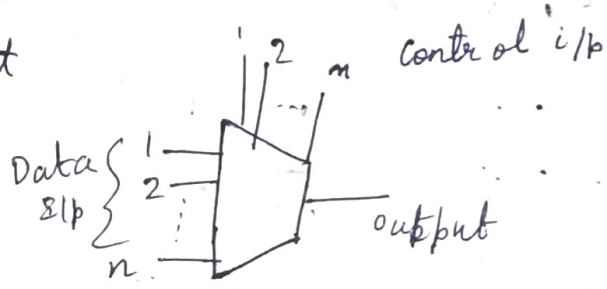
Decimal	Excess 3 code $A_3\ A_2\ A_1\ A_0$	BCD $B_3\ B_2\ B_1\ B_0$
0	0 0 0 0	x x x x
1	0 0 0 1	x x x x
2	0 0 1 0	x x x x
3	0 0 1 1	0 0 0 0
4	0 1 0 0	0 0 0 1
5	0 1 0 1	0 0 1 0
6	0 1 1 0	0 0 1 1
7	0 1 1 1	0 1 0 0
8	1 0 0 0	0 1 0 1
9	1 0 0 1	0 1 1 0
10	1 0 1 0	0 1 1 1
11	1 0 1 1	1 0 0 0
12	1 1 0 0	1 0 0 1
13	1 1 0 1	x x x x
14	1 1 1 0	x x x x
15	1 1 1 1	x x x x

Unit II : Multiplexer :

→ Mux :

Block diagrams :

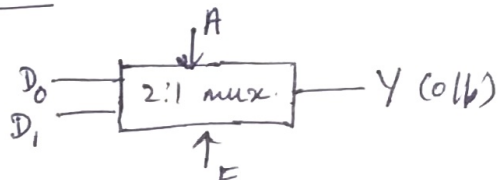




Types : 2:1, 4:1, 8:1, 16:1, 32:1

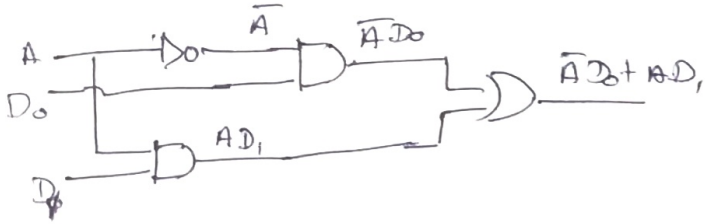
Many to one :

2:1 mux :



↳ Enable signal, works as a switch.

A	Y
0	D_0
1	D_1

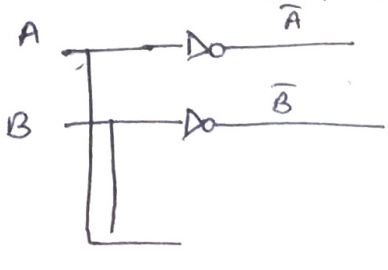


$Y = \bar{A} D_0 + A D_1$

4:1 mux :

A	B	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$\bar{A}\bar{B}D_0 + \bar{A}BD_1 + A\bar{B}D_2 + ABD_3$



$\bar{A}\bar{B}D_0 + \bar{A}BD_1 + A\bar{B}D_2 + ABD_3$

